# Wave 6 — crescent [crescent_smoke_top] | Fitness 2.867 (SPG×A×(1+N)) | SPG 3.796 N/g | Est. weight 87.5 g

Detailed plate FEM (physics::fine DKT bending + T6 membrane) with per-element thickness field (taper, grinds, relief, lattice, fuller, flutes, texture). Capacities CERTIFIED = min over 0.25 / 0.125 mm² meshes; robust = min(20 °C at −0.1 mm stock, 100 °C yield)

AESTHETIC FLOOR: PASS
H 1.46 mm · EMD 0.005 · A 0.714

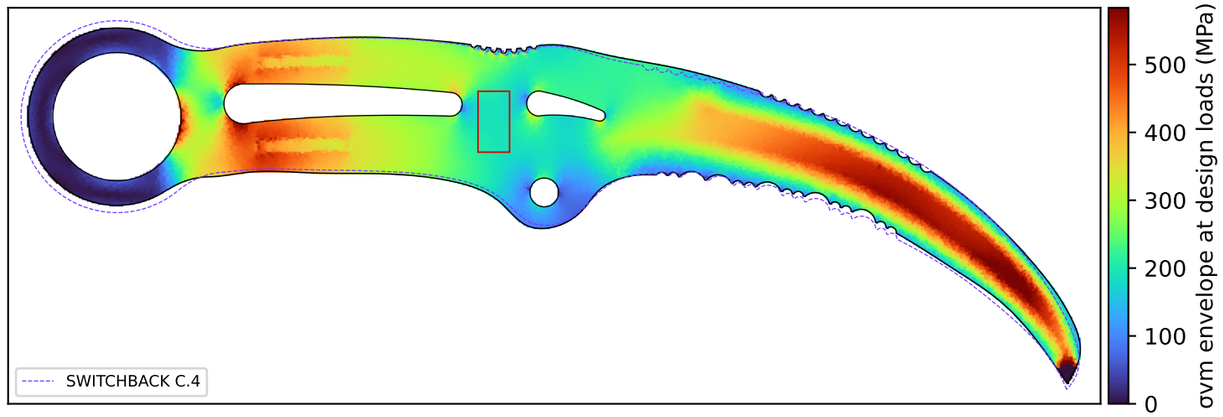
BLADE ≥ 3.375 in on all three: PASS
min 3.446 in

HARD GATES: PASS

FLOORS 180 / 260 / 1500 N (certified): PASS
lat 332 N · ring 16165 N

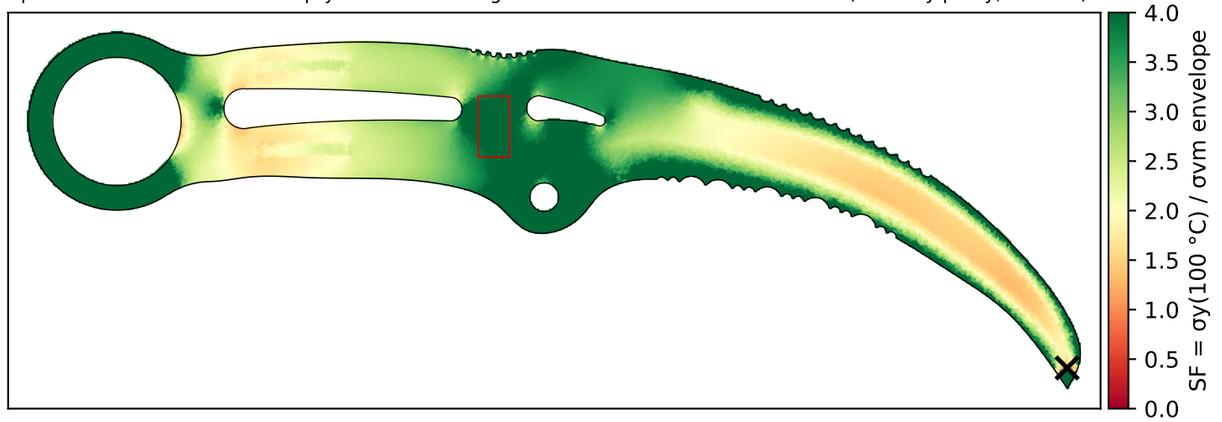
NOVELTY 0.057
(k-NN, aesthetic passers only)

von Mises envelope at design loads: lateral 260 N, pry 260 N, twist 1.26 N·m, ring pull 1500 N (4 mm tip-load zone excluded)

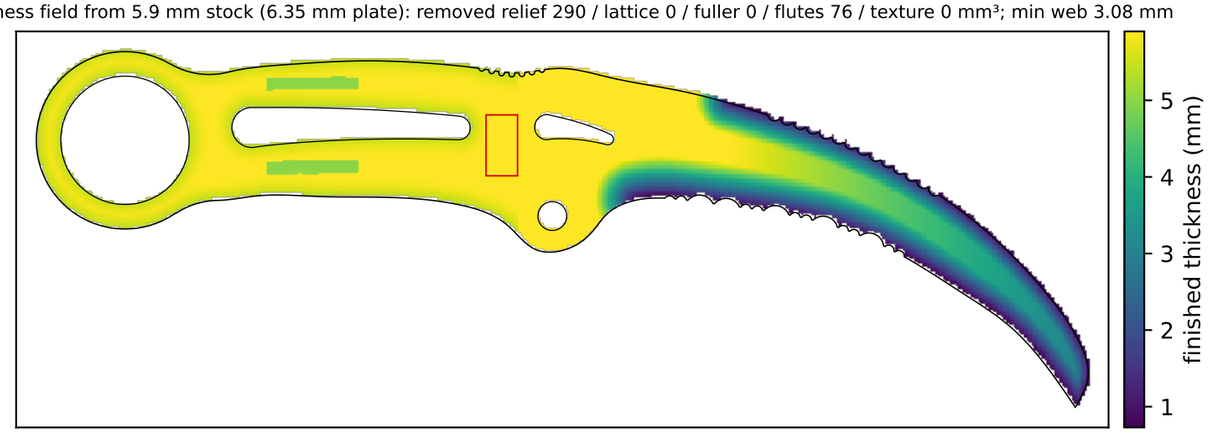



| | |
|---|---|
| tip→heel | 3.803 in (96.6) |
| horizontal reach | 3.446 in (87.5) |
| edge arc inside / spine | 3.986 / 3.947 in |
| grip / stock / Ti | 61.3 mm / 5.9 mm / Ti-6Al-4V |
| weight (finished / blank) | 87.5 g / 104.8 g |
| lateral 20 °C / robust | 364 / 332 N |
| SF hiker 180 / hunter 260 | 1.84 / 1.28 |
| ring pull 20 °C / robust (SF) | 17725 / 16165 N (10.8) |
| tip pry 20 °C / twist 20 °C | 712 N / 1.25 N·m |
| SF 100 °C lat / pry / ring / twist* | 1.28 / 3.03 / 12.00 / 0.93 |
| governing lateral at | (177, -33) mm |
| holder box (oriented) | 198.8 × 99.2 mm @ -9.0° |
| aesthetic A / H / EMD | 0.714 / 1.46 mm / 0.005 |
| straight run / min blend R | 11.5 / 9.3 mm |
| novelty / SPG | 0.057 / 3.796 N/g |

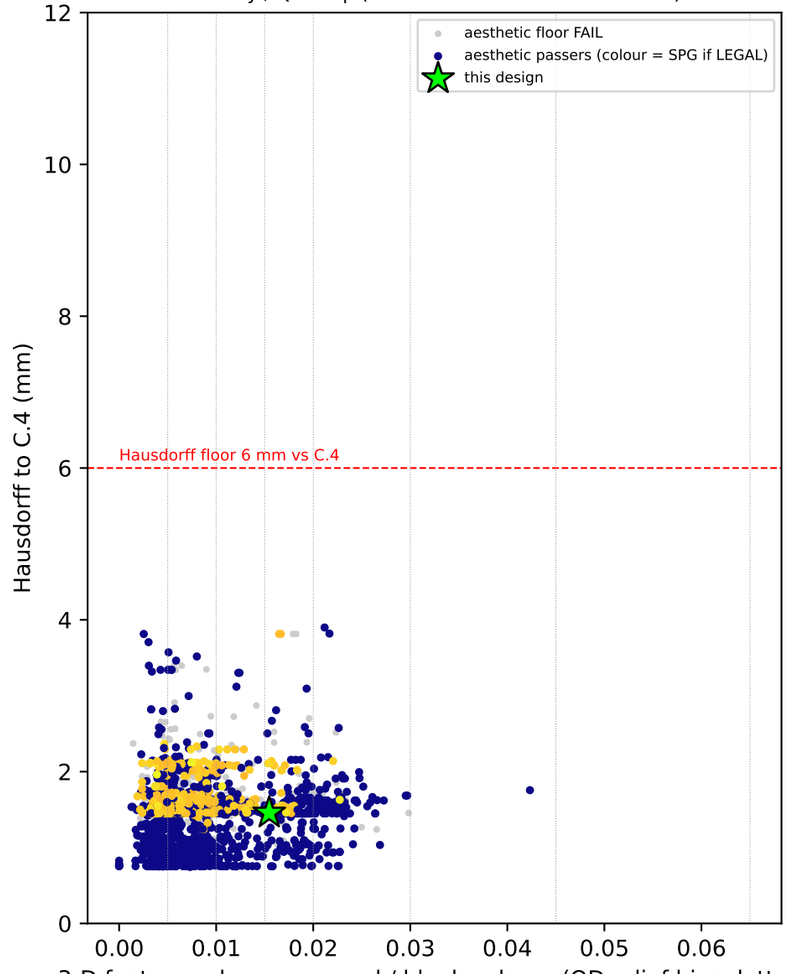
safety factor at 100 °C, envelope of all four cases (min 0.93 at x=188, y=-49 mm)
per case: lateral 260 N 1.28 · pry 260 N 3.03 · ring 1500 N 12.00 · twist 1.26 N·m 0.93 (advisory proxy, no floor)




thickness field from 5.9 mm stock (6.35 mm plate): removed relief 290 / lattice 0 / fuller 0 / flutes 76 / texture 0 mm³; min web 3.08 mm




novelty / QD map (all evaluations of the smoke run)




source out/crescent_smoke | author Jepson Taylor | generator ops/wave_figures/make_crescent_plate.py | utility / hiking karambit only